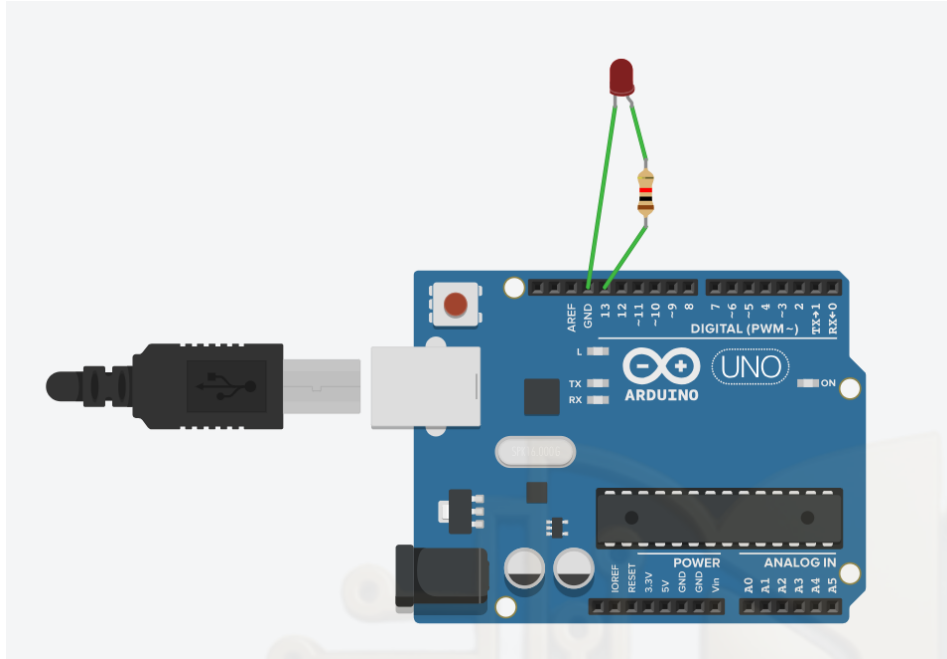


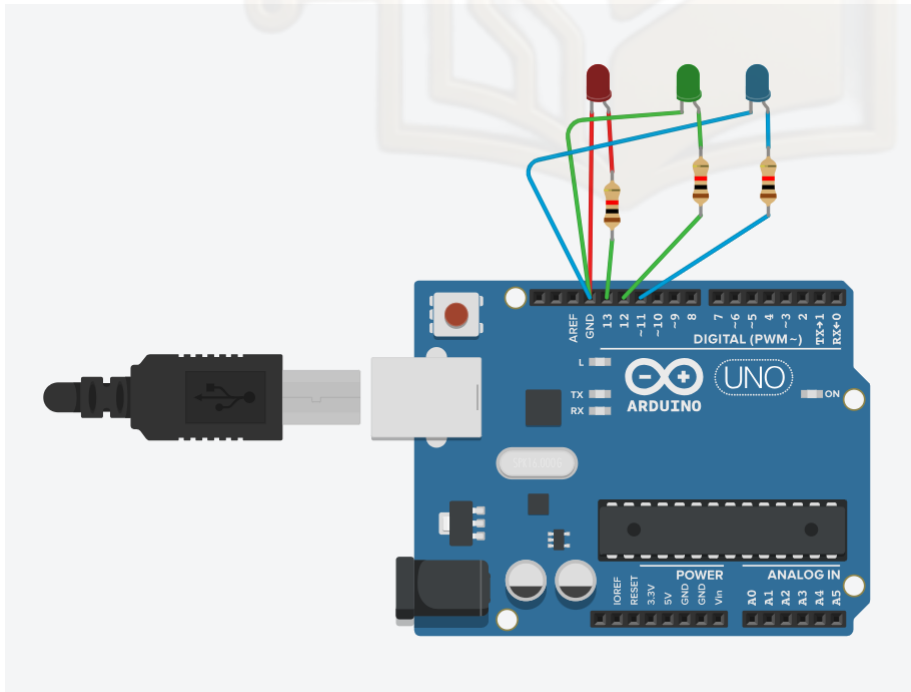
IOT Projects Using Arduino

1. Simple LED on and Off

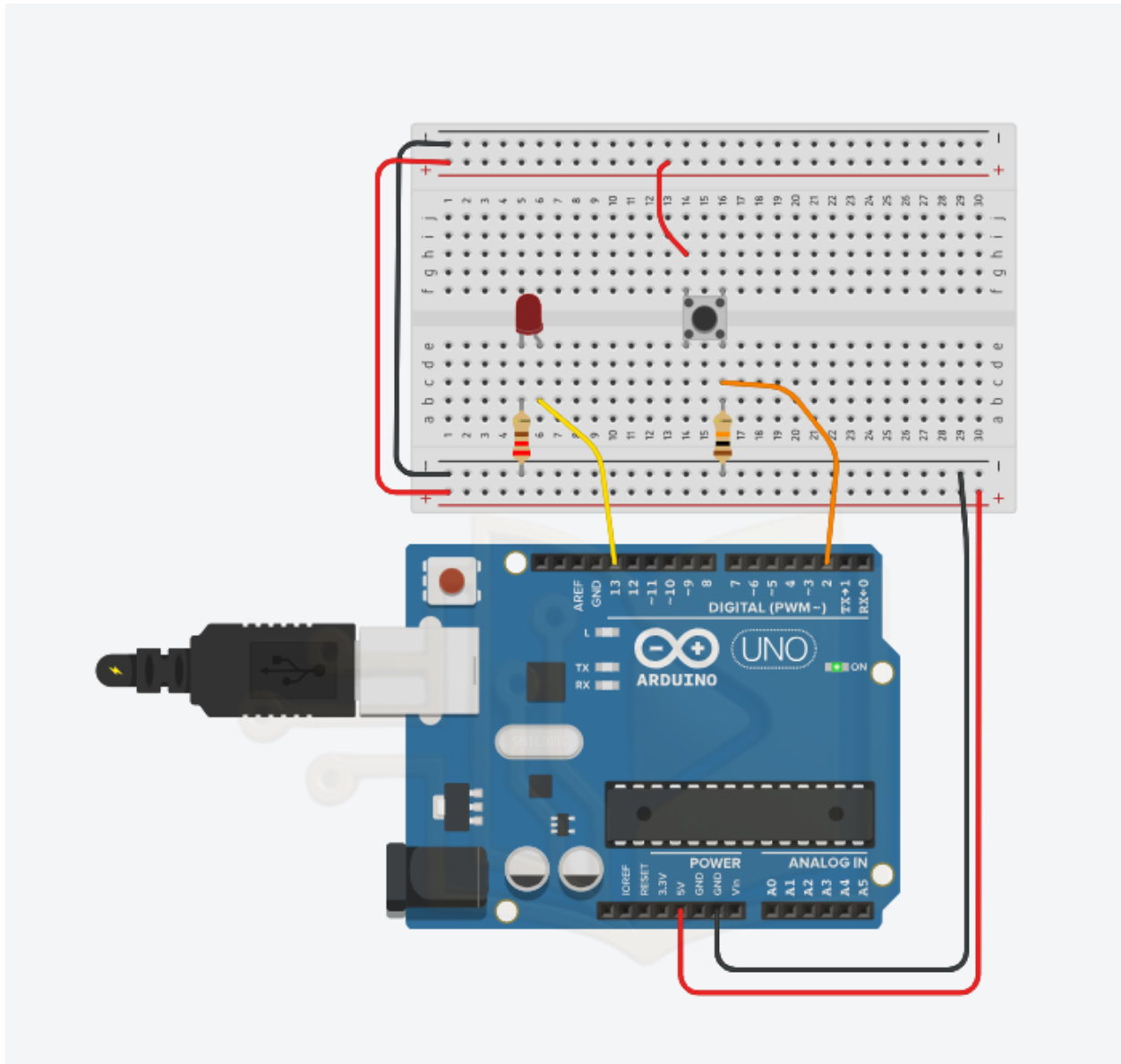
Code Circuit Configuration

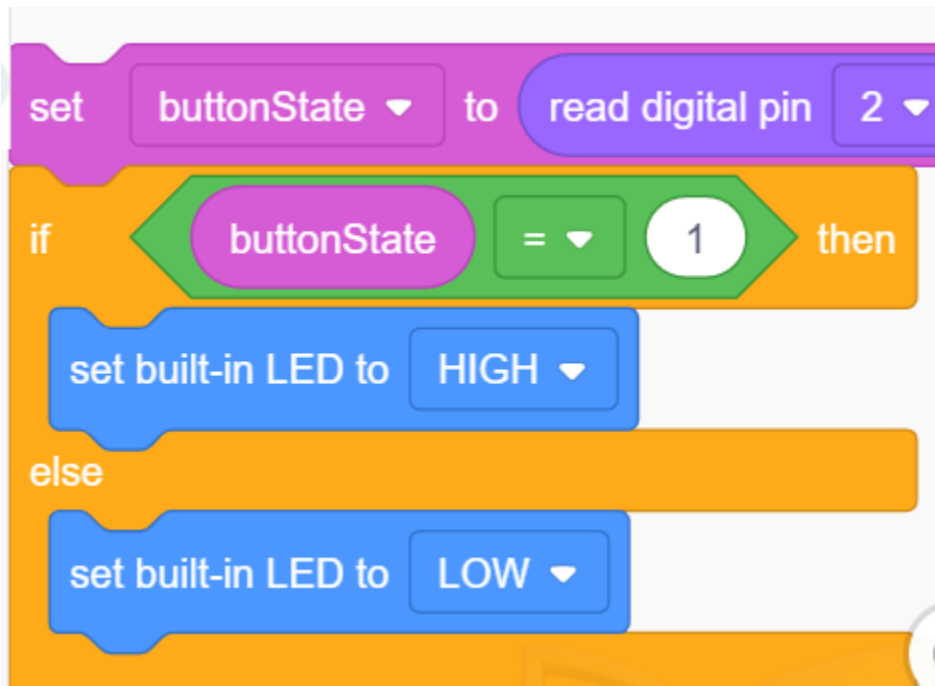


2. 3 led RGB with a delay of 2s

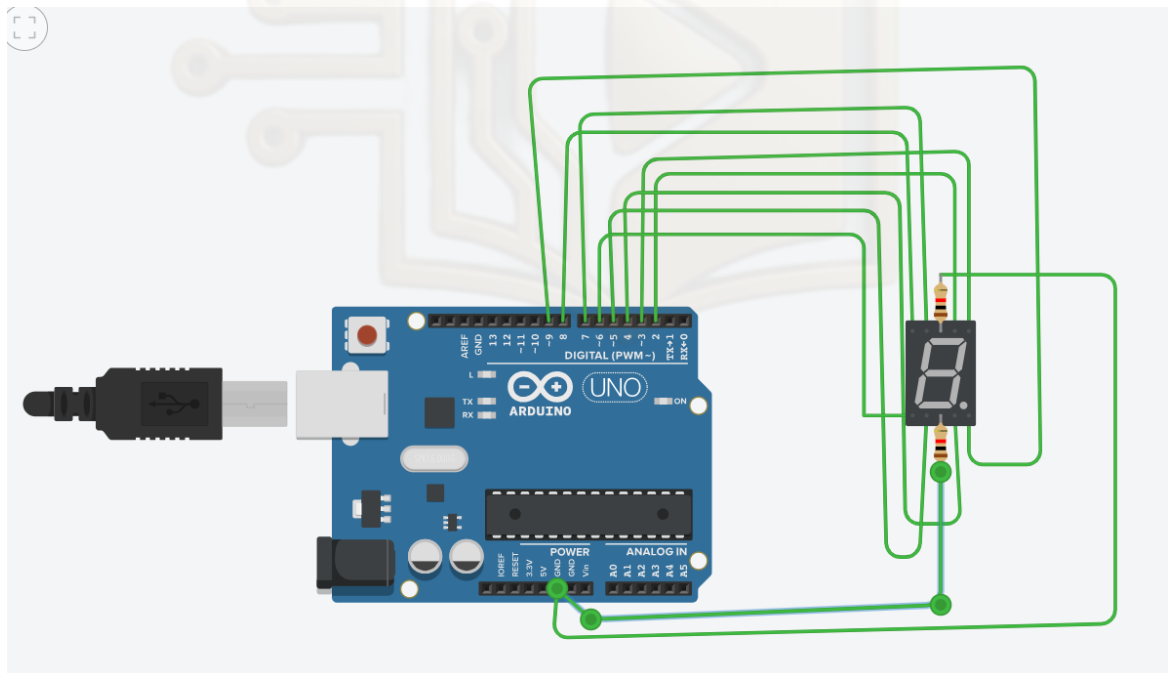


3. Led On and Off with push button

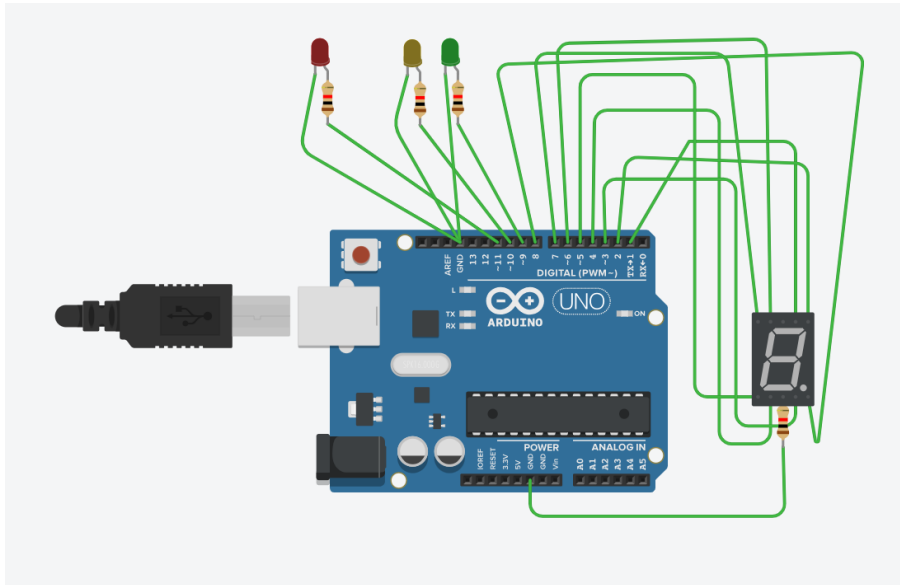




4. 7-Segment Display



5. 7-segment display with LED traffic light



Program

```
int count = 5;
// a b c d e f g
int segPins[7] = {1,2, 3, 4, 5, 6, 7};
// digits 0-5
byte digits[6][7] = {
  {1,1,1,1,1,1,0}, // 0
  {0,1,1,0,0,0,0}, // 1
  {1,1,0,1,1,0,1}, // 2
  {1,1,1,1,0,0,1}, // 3
  {0,1,1,0,0,1,1}, // 4
  {1,0,1,1,0,1,1} // 5
};
void setup() {
  pinMode(LED_BUILTIN, OUTPUT);
  for (int i = 0; i < 7; i++) {
    pinMode(segPins[i], OUTPUT);
  }
}
void displayDigit(int num) {
  for (int i = 0; i < 7; i++) {
    digitalWrite(segPins[i], digits[num][i]);
  }
}
void loop() {
  displayDigit(count);
  digitalWrite(LED_BUILTIN, HIGH);
```

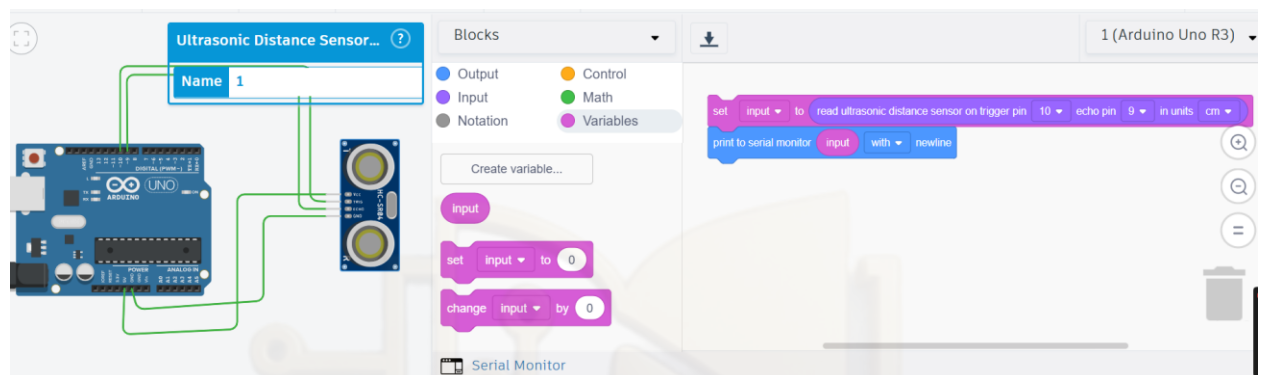
```

delay(5000);
digitalWrite(LED_BUILTIN, LOW);
delay(2000);
count--;
if (count < 0) {
  count = 5;
}
}

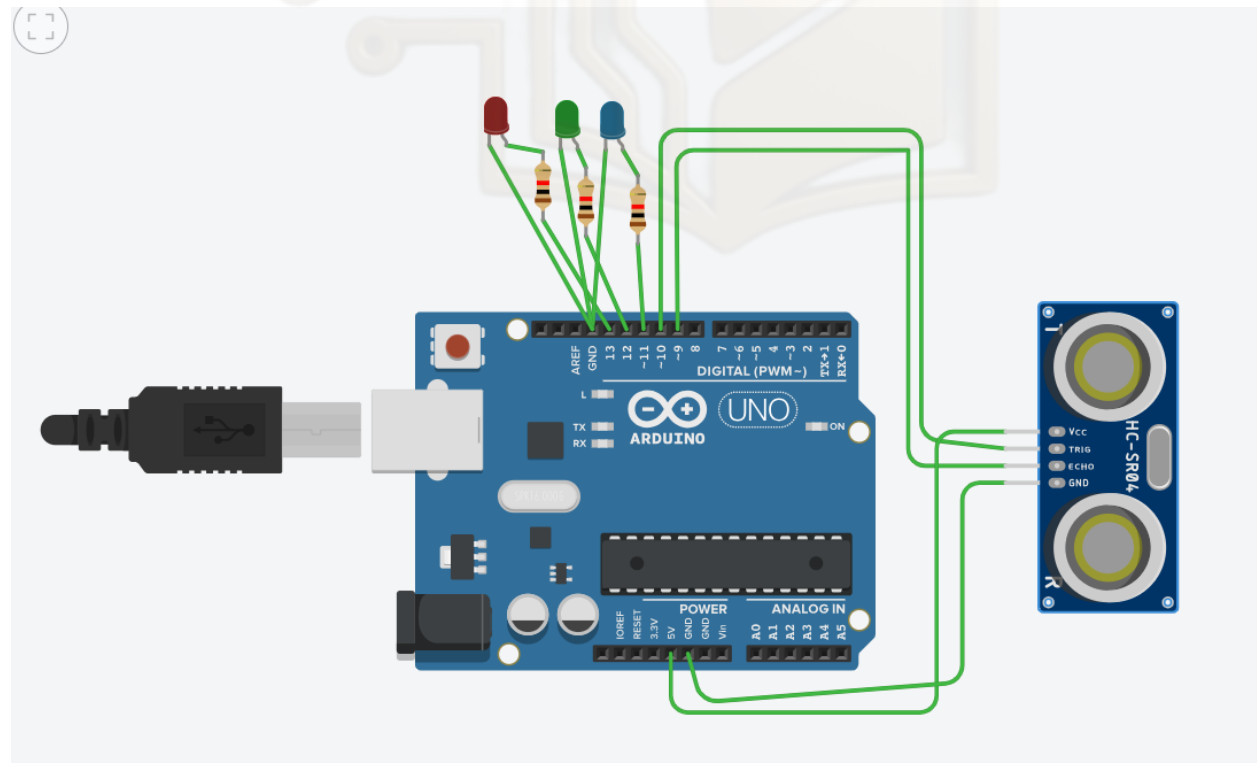
```

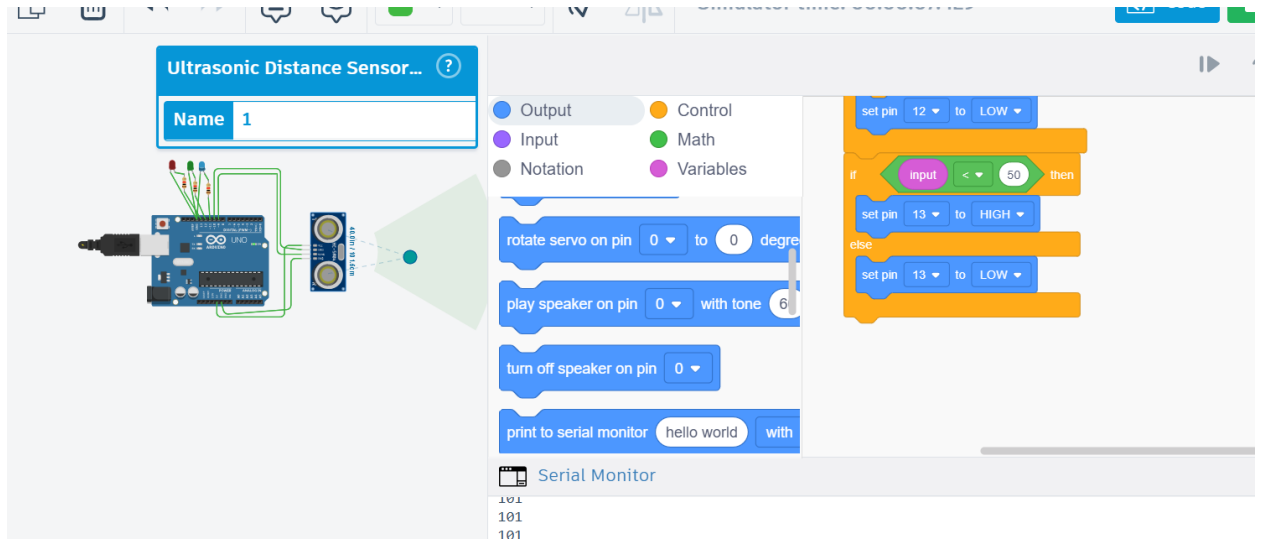
6. Ultrasonic Sensor Interfacing

Create an input variable and attach

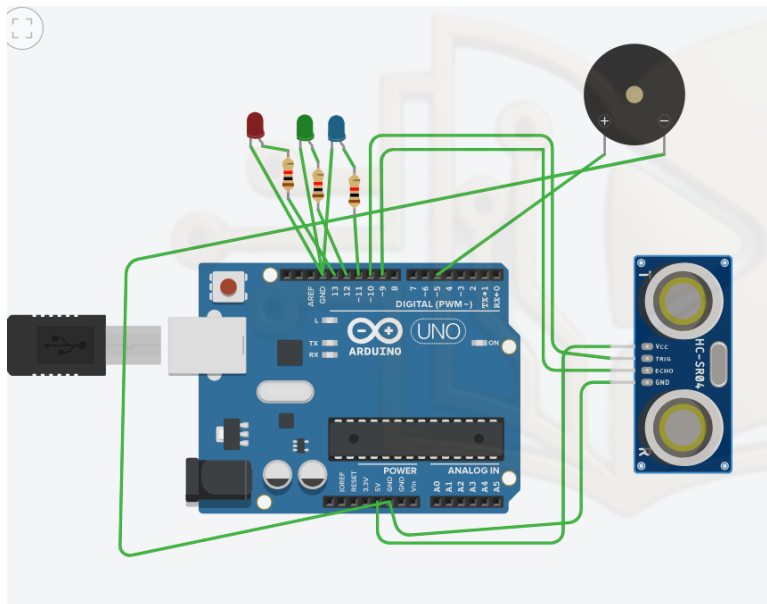


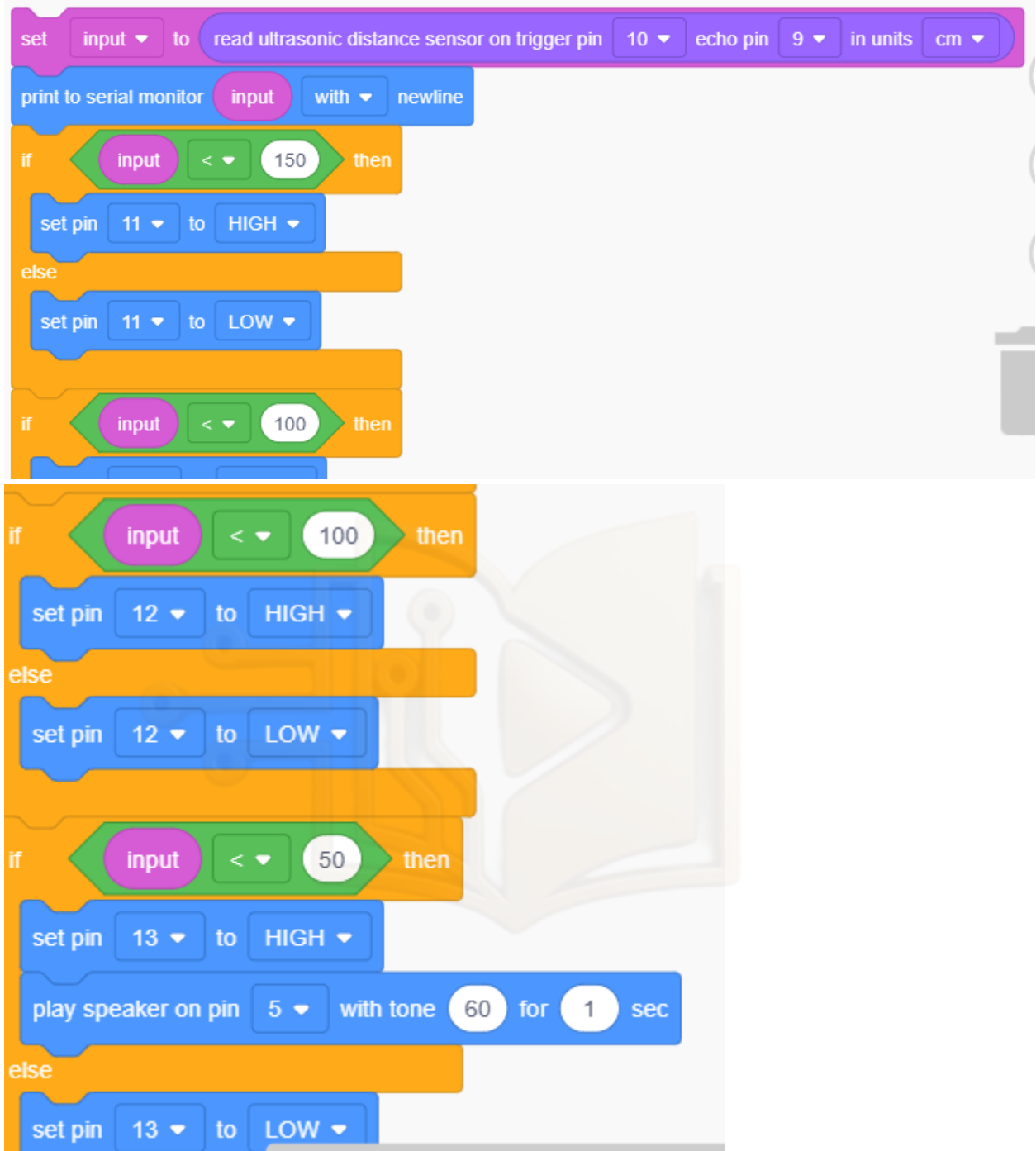
7. Distance Measurement With LED & Arduino and Ultrasonic sensors



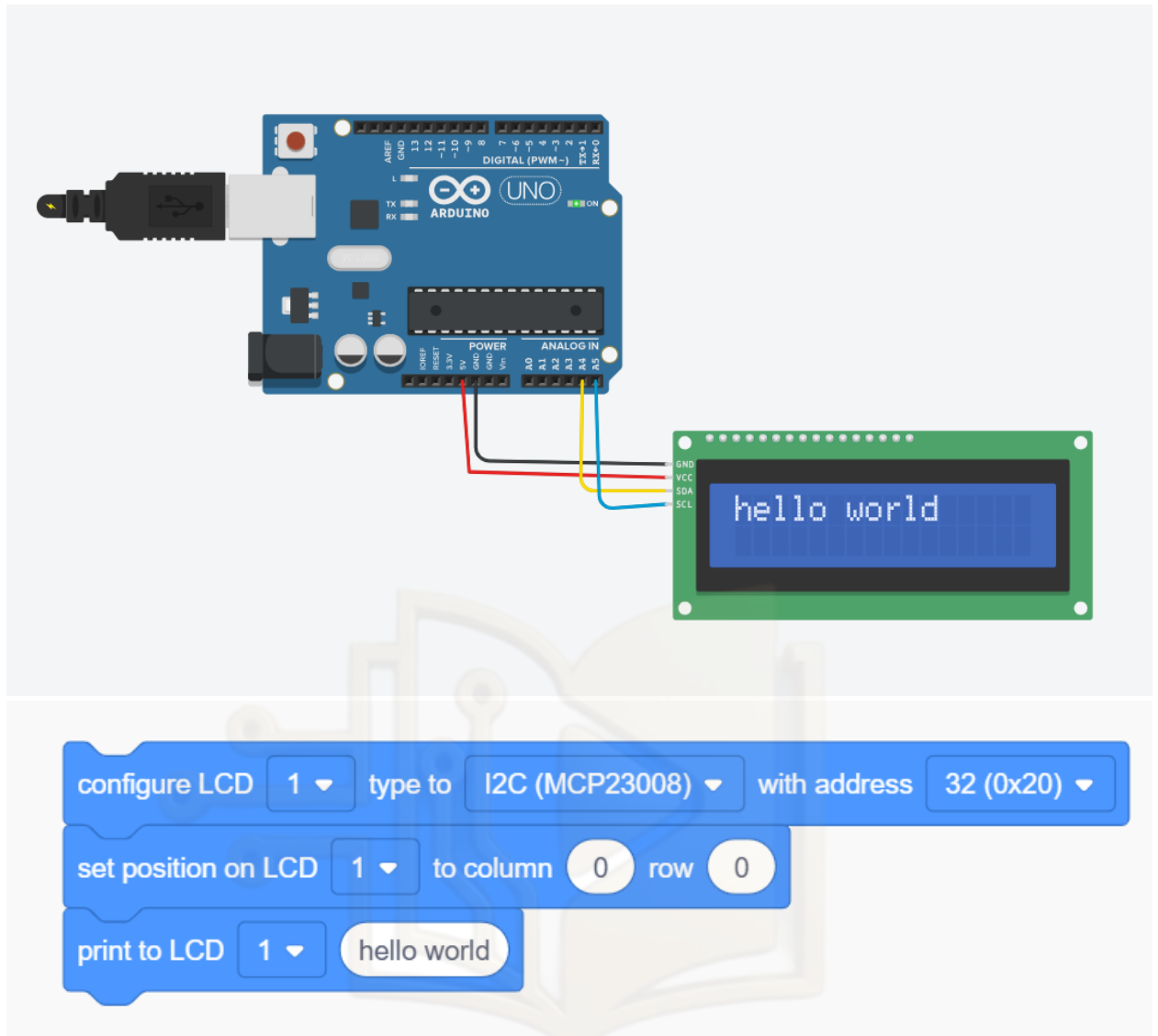


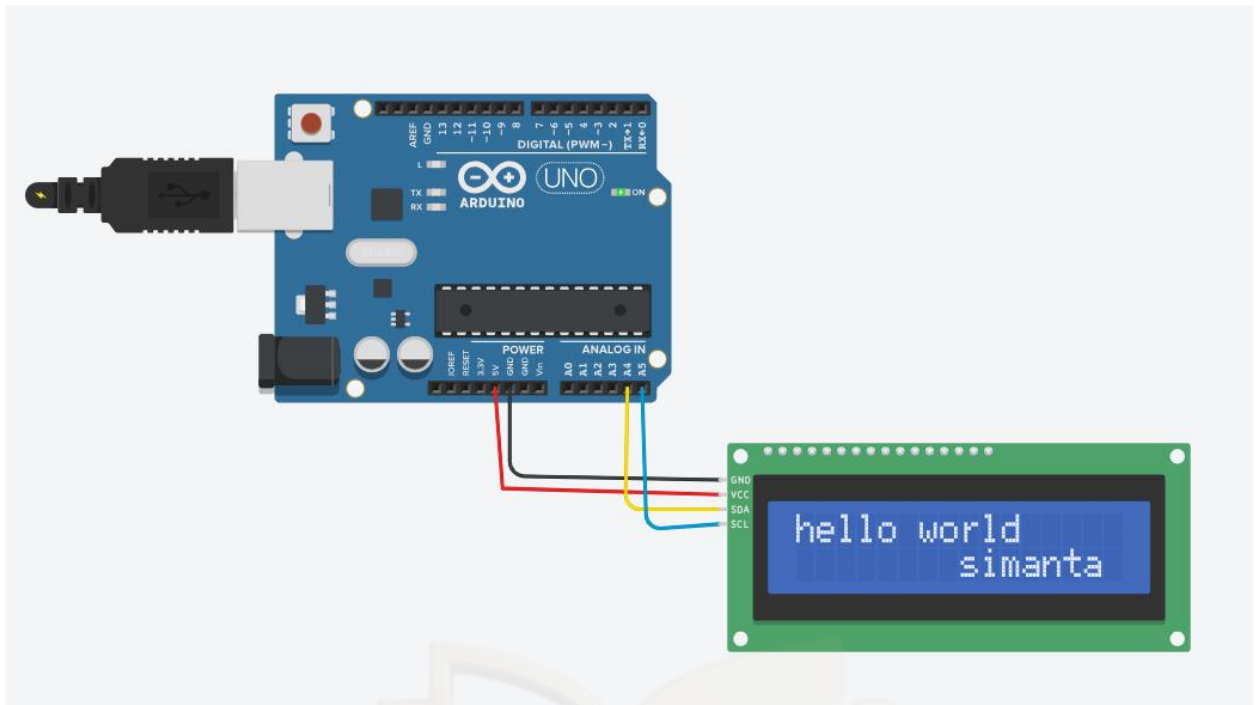
8. With an alarm as well

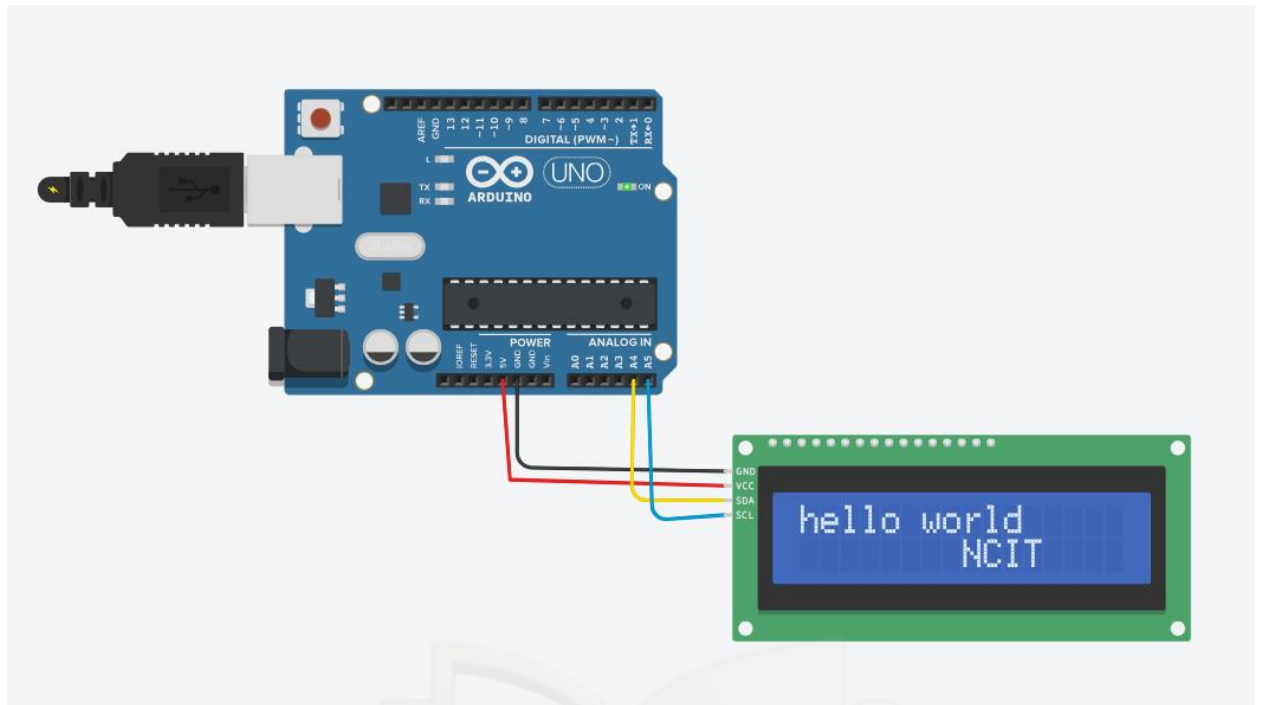




9. Print Hello world

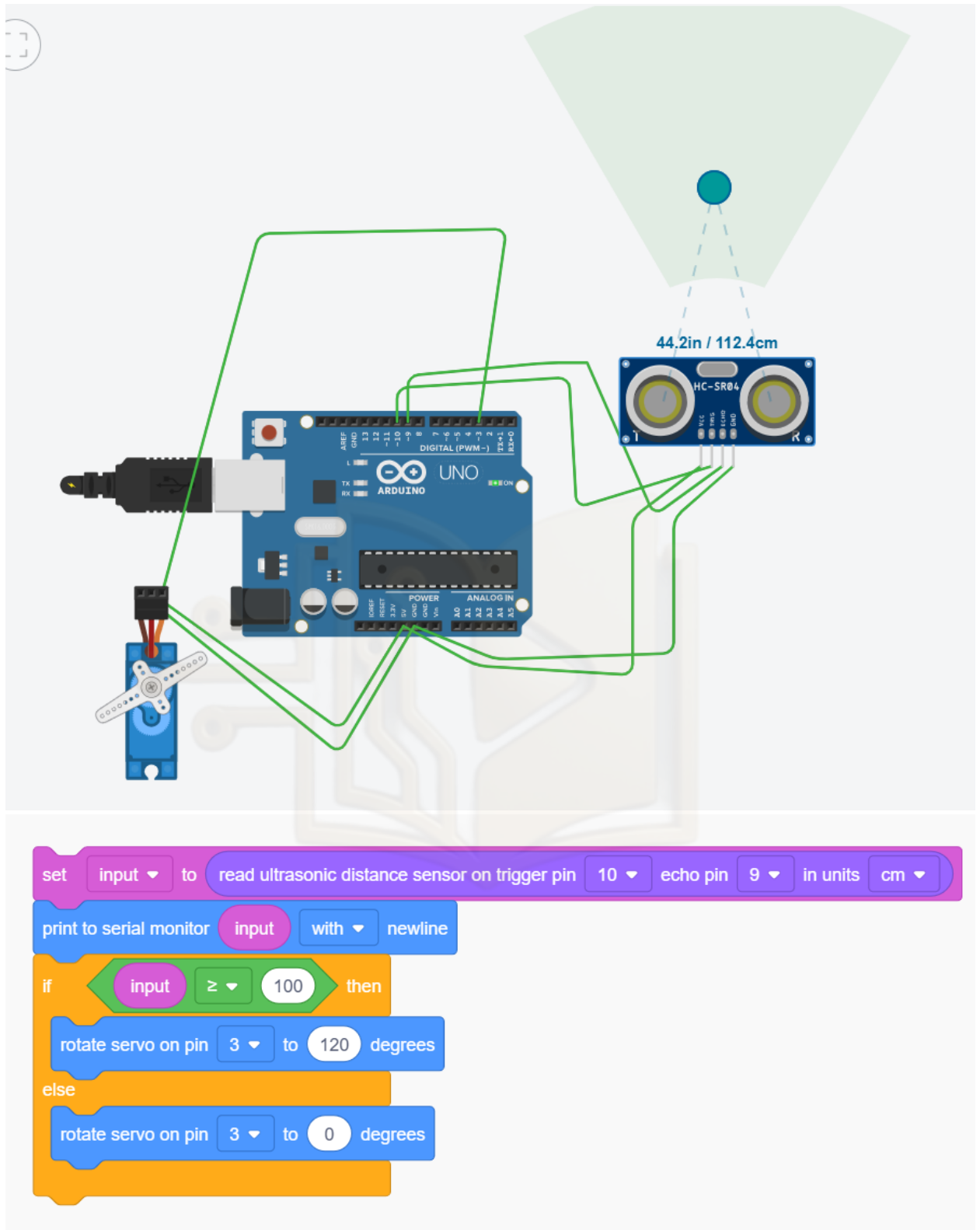




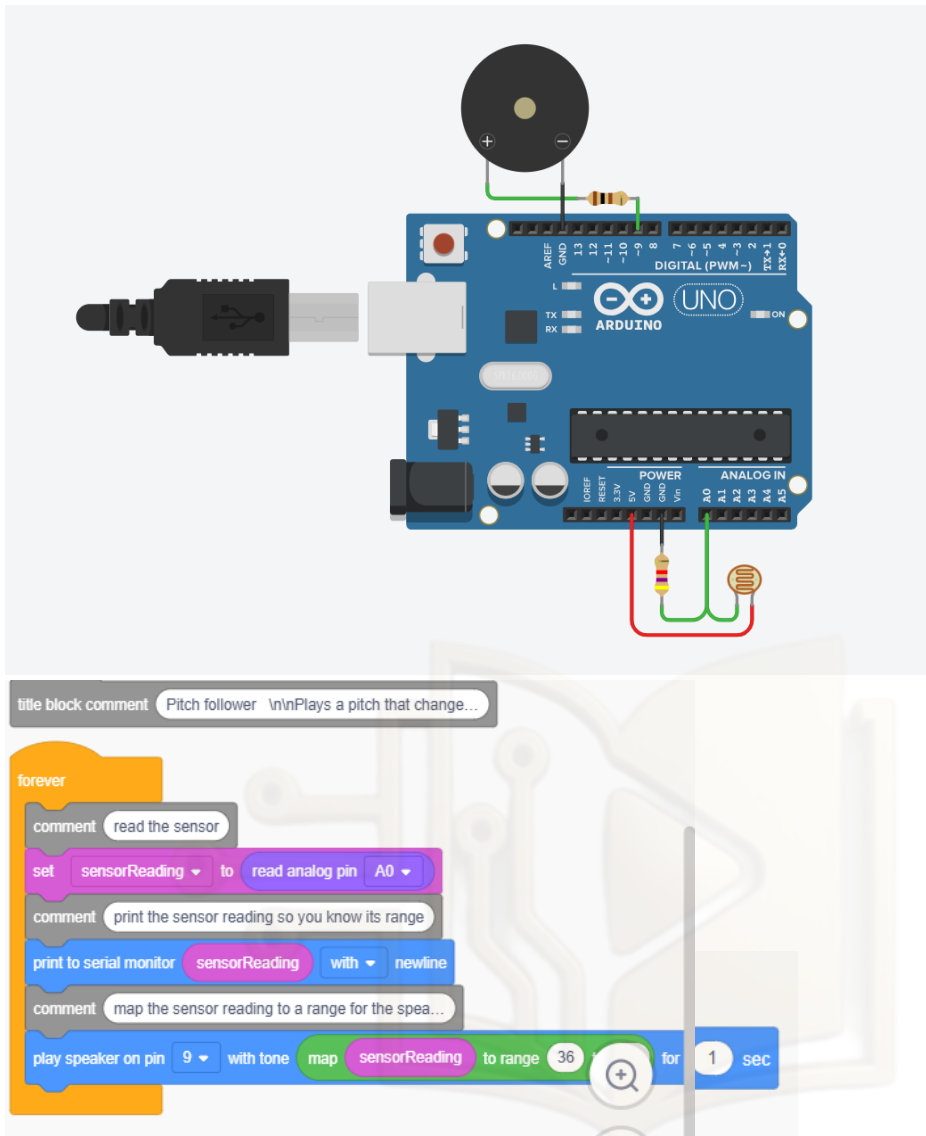




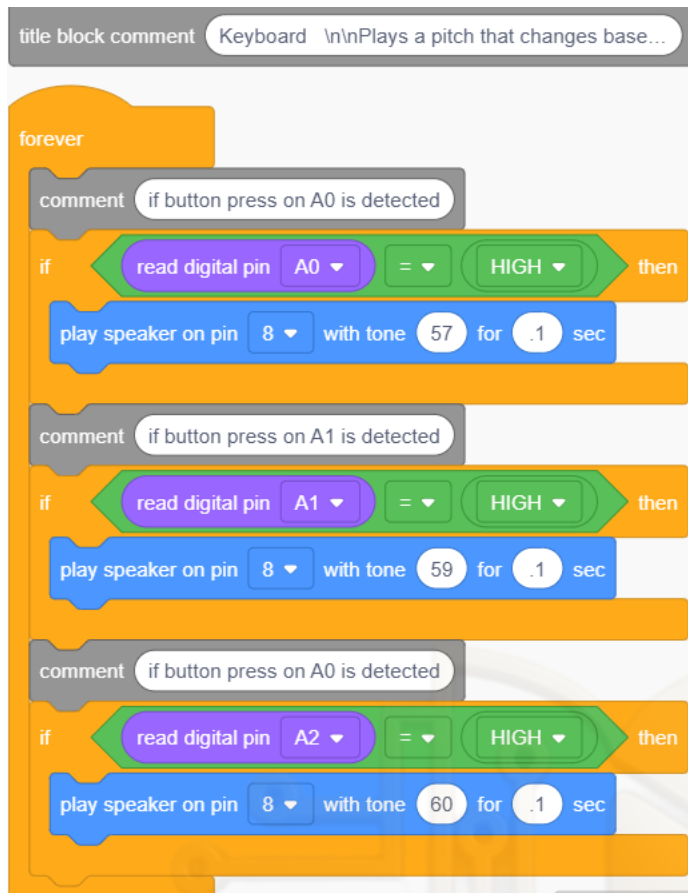
10. Ultrasonic sensor with servo motor



11. Tone Pitch follower



12. Tone Keyword



13. Interfacing Temperature Sensor with Arduino using Tinker CAD

```

// Fire Alarm System with Gas + Temperature + Buzzer + LED

const int gasPin = A0;    // MQ2 Smoke/Gas Sensor
const int tempPin = A1;   // Temperature sensor (LM35/TMP36)
const int buzzer = 7;     // Buzzer
const int led = 13;       // LED
int gasThreshold = 300;   // Gas threshold (adjust after testing)
float tempThreshold = 50.0; // Celsius threshold for fire

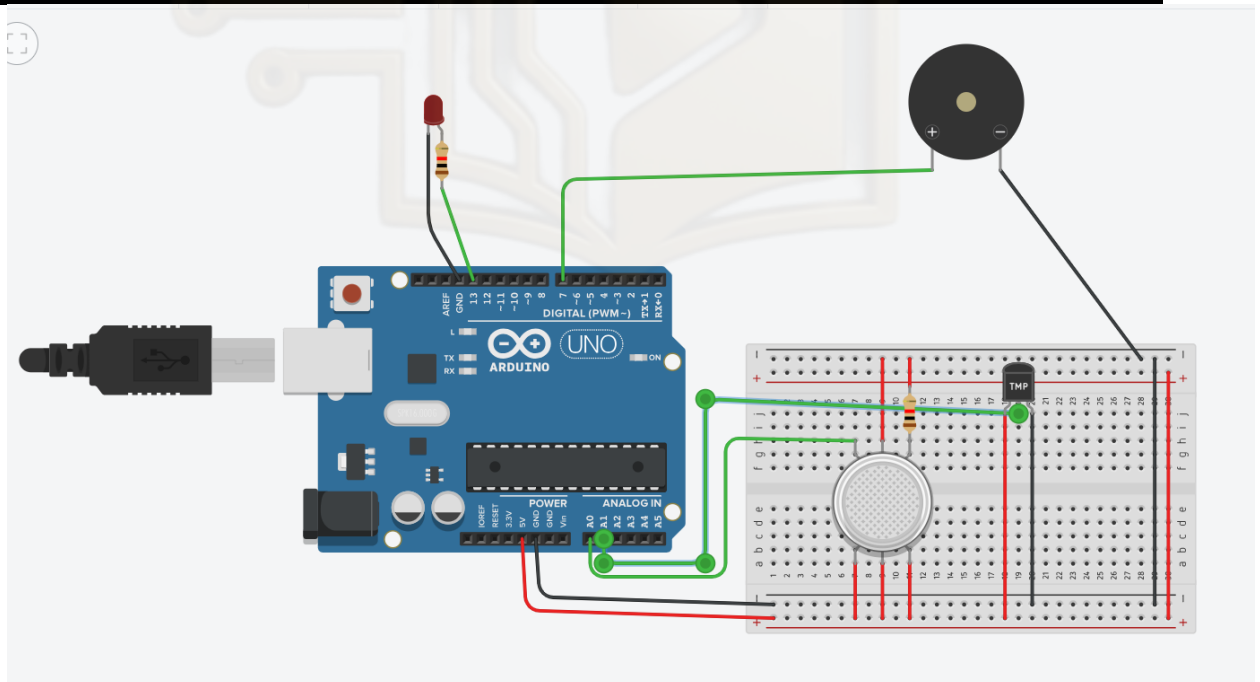
void setup() {
  pinMode(buzzer, OUTPUT);
  pinMode(led, OUTPUT);
  Serial.begin(9600);
  Serial.println("Fire Alarm System Ready");
}

void loop() {
  // Read Gas
  int gasValue = analogRead(gasPin);
  // Read Temp
  int tempValue = analogRead(tempPin);
  
```

```

float voltage = tempValue * (5.0 / 1023.0);
float tempC = (voltage - 0.5) * 100.0; // TMP36 formula (if LM35 use voltage*100)
// Serial output
Serial.print("Gas Value = ");
Serial.print(gasValue);
Serial.print(" | Temp = ");
Serial.print(tempC);
Serial.println(" °C");
// Alarm condition
if (gasValue > gasThreshold || tempC > tempThreshold) {
  digitalWrite(led, HIGH);
  digitalWrite(buzzer, HIGH);
  delay(200);
  digitalWrite(buzzer, LOW);
  delay(200);
}
else {
  digitalWrite(led, LOW);
  digitalWrite(buzzer, LOW);
}
delay(500);
}

```



14. Controlling LEDs using an IR Remote

```

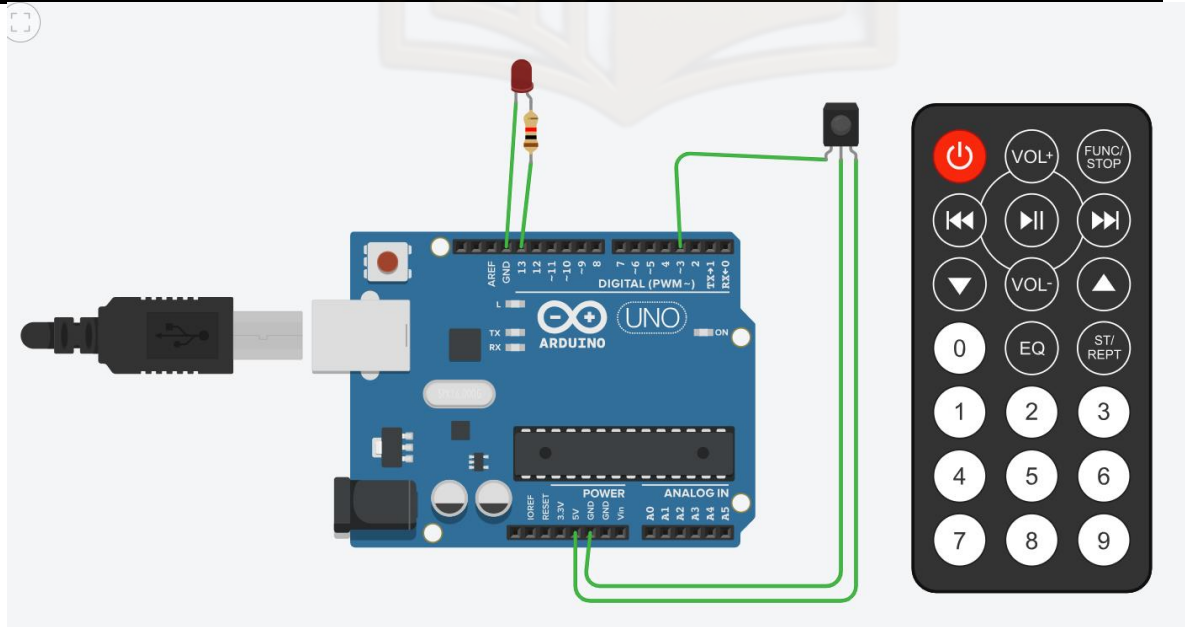
#include <IRremote.h>
const int rcvPin = 3;

```

```

const int ledPin = 5;
void setup() {
  Serial.begin(9600);
  IrReceiver.begin(rcvPin, ENABLE_LED_FEEDBACK); // Correct initialization
  pinMode(ledPin, OUTPUT);
  Serial.println("IR Receiver Ready");
}
void loop() {
  if (IrReceiver.decode()) {
    unsigned long value = IrReceiver.decodedIRData.decodedRawData;
    Serial.println(value);
    switch (value) {
      case 4010852096: // Button 1
        Serial.println("Button 1 Pressed");
        digitalWrite(ledPin, HIGH);
        break;
      case 3994140416: // Button 2
        Serial.println("Button 2 Pressed");
        digitalWrite(ledPin, LOW);
        break;
      default:
        Serial.println("Unknown Button");
    }
    IrReceiver.resume(); // IMPORTANT
  }
}

```

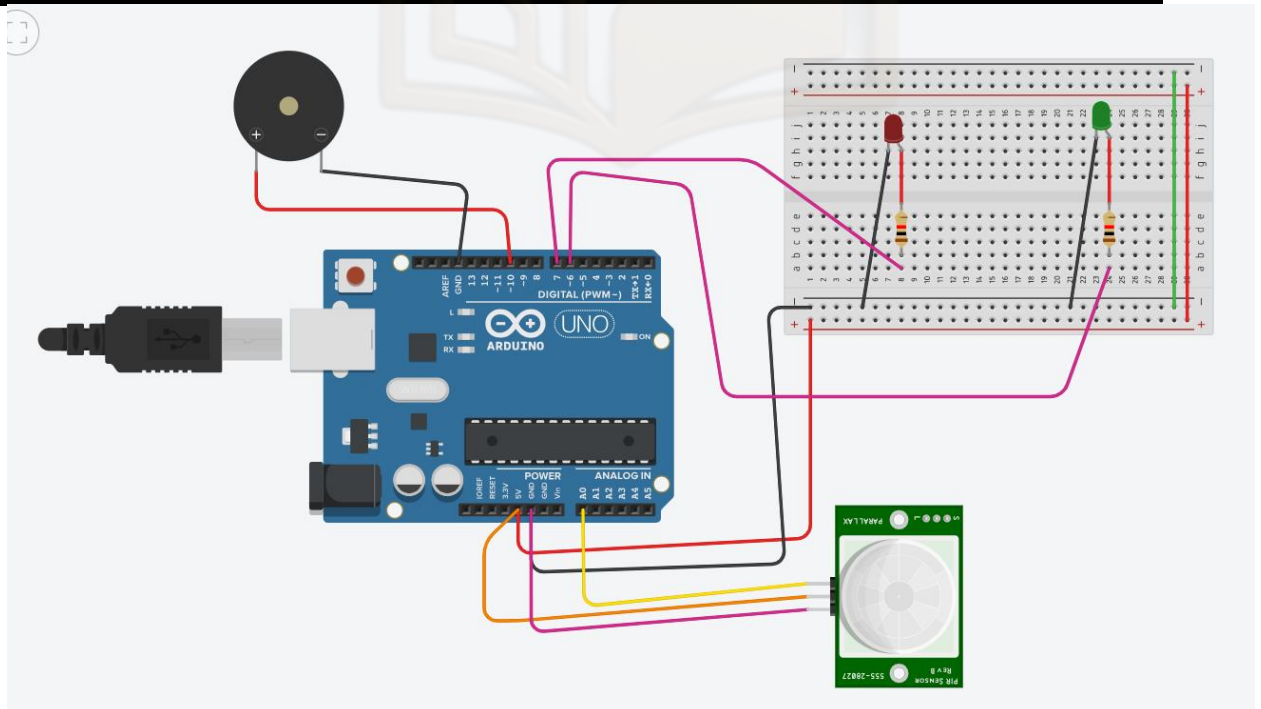


15. PIR motion sensor

```

int pirsensor = 0;
void setup()
{
  pinMode(A0, INPUT);
  Serial.begin(9600);
  pinMode(7, OUTPUT);
  pinMode(6, OUTPUT);
  pinMode(10, OUTPUT);
  pinMode(10, OUTPUT);
}
void loop()
{
  pirsensor = analogRead(A0);
  Serial.println(pirsensor);
  if (pirsensor >= 100) {
    digitalWrite(7, LOW);
    digitalWrite(6, HIGH);
    tone(10, 92, 100); // play tone 30 (F#2 = 92 Hz)
  } else {
    digitalWrite(7, HIGH);
    digitalWrite(6, LOW);
    digitalWrite(10, LOW);
  }
  delay(10); // Delay a little bit to improve simulation performance
}

```



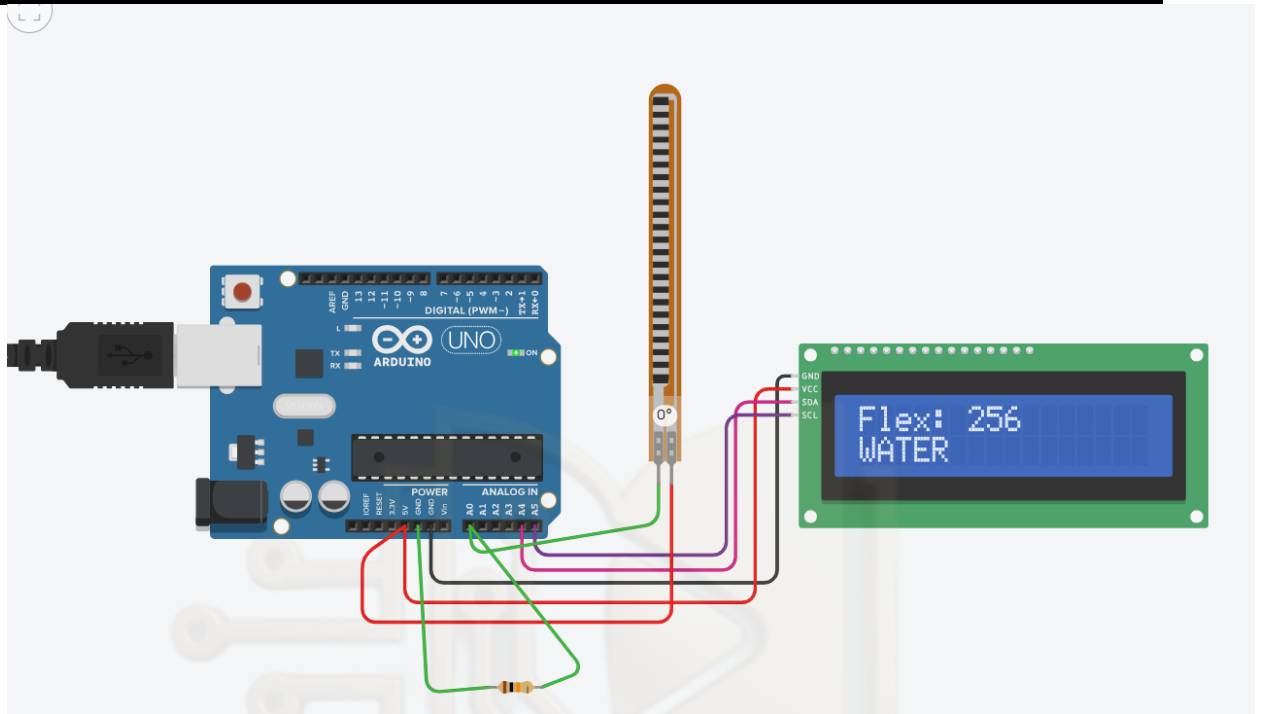
16. Flex sensor for sign language

```
#include <Adafruit_LiquidCrystal.h>
Adafruit_LiquidCrystal lcd(0);
const int flexPin = A0;
int stableValue = 0;
void setup() {
  Serial.begin(9600);
  lcd.begin(16, 2);
  lcd.clear();
  // Force reference stability
  analogReference(DEFAULT);
  // Warm up ADC
  for (int i = 0; i < 50; i++) {
    analogRead(flexPin);
    delay(10);
  }
  stableValue = analogRead(flexPin);
}
int readStableFlex() {
  long sum = 0;
  for (int i = 0; i < 20; i++) {
    sum += analogRead(flexPin);
    delay(5);
  }
  return sum / 20;
}
void loop() {
  int newValue = readStableFlex();
  // HARD deadband (prevents self-change)
  if (abs(newValue - stableValue) > 15) {
    stableValue = newValue;
  }
  Serial.print("Flex: ");
  Serial.println(stableValue);
  lcd.setCursor(0, 0);
  lcd.print("Flex: ");
  lcd.print(stableValue);
  lcd.print(" ");
  lcd.setCursor(0, 1);
  if (stableValue < 50) lcd.print("NO SENSOR  ");
  else if (stableValue < 200) lcd.print("HAND OPEN  ");
  else if (stableValue < 400) lcd.print("WATER      ");
}
```

```

else if (stableValue < 600) lcd.print("FOOD    ");
else if (stableValue < 800) lcd.print("BATHROOM ");
else lcd.print("HELP    ");
delay(200);
}

```



Do Yourself

- 17. Password-Protected Security System project using Arduino, Keypad, Piezo Buzzer, and LED**
- 18. Voting system with LCD**
- 19. Light Intensity Measurement using LDR sensor**
- 20. Vibration sensor**